

Yijin Zeng

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SUMMARY

Final year PhD candidate in Statistics and Machine Learning at Imperial College London.

- Research focuses on developing new methodologies for statistical hypothesis testing, statistical correlation, and change point detection in the presence of missing data.
- Applications include A/B testing, causal inference, information retrieval, and continual learning.
- Strong statistics and machine learning theoretical foundations with hands on experience in end-to-end data science, machine learning, and deep learning projects.
- Proficient in Python and R. Solid understanding of data structures and algorithms.

EDUCATION

- **Imperial College London**, London, UK Oct. 2021– Oct. 2025 (Estimated)
PhD in Statistics and Machine Learning ([StatML CDT](#), Imperial & University of Oxford)
– *Funded by Roth Scholarship*
- **Imperial College London**, London, UK Oct. 2020 – Sep. 2021
MSc in Statistics
– *Degree classification: Distinction*
- **Beijing Institute of Technology**, Beijing, China Sep. 2016 – Jun. 2020
BSc in Mathematics
– *Top 10% of cohort; Thesis with Distinction*

WORK EXPERIENCE

- **TESCO Data Scientist PhD Internship (London)** Aug.2024 – Oct.2024
– Worked with forecasting team and supply chain team, delivering an end-to-end machine learning pipeline for demand forecasting, including feature selections, feature engineering, data cleaning, pre-processing, model training, and evaluation.
- Designed and trained multiple deep neural network architectures for supply chain forecasting, achieving accuracy comparable or higher (5% ~ 10% depending on the product categories) to TESCO's current approach with less workflows.
- **IQVIA Data Analyst Internship (Beijing)** Jul.2019 – Sep.2019
– Cleaned and analyzed survey and experimental datasets, performing feature engineering, missing-data handling, and statistical testing for supporting final conclusions.
- Worked on data visualization for efficiently communicating numerical results to managers and non-technical stakeholders.

PUBLICATIONS

- **Zeng Y**, Adams NM, Bodenham DA. On two-sample testing for data with arbitrarily missing values. arXiv preprint arXiv:2403.15327. 2024 Mar 22.
– *A two-sample hypothesis testing method for partially observed univariate data to detect location shifts, with no assumption of the values of missing data.*
- **Zeng Y**, Adams NM, Bodenham DA. MMD Two-sample Testing in the Presence of Arbitrarily Missing

Data. arXiv preprint arXiv:2405.15531. 2024 May 24.

– *Extended our framework for high-dimensional data with any distributional shifts.*

- **Zeng Y**, Adams NM, Bodenham DA. Exact Bounds of Spearman’s footrule in the Presence of Missing Data with Applications to Independence Testing. arXiv preprint arXiv:2501.11696. 2025 Jan 20.
 - *Tight bounds for Spearman’s footrule in the presence of missing data, with applications for independence testing.*

OPEN SOURCE PACKAGES

- **Zeng Y**, Bodenham D, and Adams N. Package ‘wmwm’.
 - *R and Python packages for performing Wilcoxon-Mann-Whitney two sample hypothesis testing in the presence of missing data.*
- **Zeng Y**. Package ‘bosfr’.
 - *R package for computing tight bounds of Spearman’s footrule in the presence of missing data.*

SELECTED DATA SCIENCE PROJECTS

- [E-Commerce Order Cancellation Prediction](#)
 - *Predict order cancellation with highly imbalanced dataset for real-world impact.*
- [Deep Learning Hangman Game](#)
 - *Play word guessing games using deep learning with high success rate (~60%).*
- [Stock Clustering](#)
 - *Cluster stocks using daily return to produce balanced investment groups.*

SELECTED ADVANCED COURSES

- Machine Learning in Finance (by Dr. Mikko Pakkanen)
 - *Machine learning algorithms for high-frequency trading (HFT)*
- Computational Optimal Transport and Statistical Learning with Missing Values (by Prof. Gabriel Peyré & Dr. Julie Josse)
 - *Optimal transport for machine learning; dealing with missing data in machine learning*
- Selective Inference (by Dr. Daniel Garcia Rasines & Prof. Alastair Young)
 - *Valid inference for the ‘post selection’ problem in modern statistics and machine learning*

SELECTED TEACHING EXPERIENCE

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| • Probability for Statistics (MSc) | Summer term 2023 |
| • Data Science (MSc) | Spring term 2024 |
| • Statistical Clinics (MSc) | Summer term 2024 |
| • Probability and Statistics (BSc) | Spring term 2024 |

SKILLS

- Language: Chinese (Native), English (Proficient).
- Programming & Tools: Python, R, TensorFlow, pandas, NumPy, scikit-learn, MATLAB; Azure ML; HPC; Linux; Git.